Homework 09 Two Way ANOVA / ANCOVA / GLM

Due by 11:59pm, Friday, November 21, 2025, 11:59pm

S&DS 2300/5300/ENV 757

This assignment uses data from the International Social Survey Program on Environment from 2020 There are 224 questions from 21718 individuals across 14 countries.

The data you'll need is here (https://raw.githubusercontent.com/jreuning/sds230_data/refs/heads/main/ISS_2020.csv). Be aware that it will take a few moments to load this data.

You'll also want the codebook that describes the variables (https://raw.githubusercontent.com/jreuning/sds230_data/refs/heads/main/ISS_2020_Codebook.pdf). The final pages of the codebook have information on how to translate variable names to questions in the survey.

1) Data Set creation *(23 pts - 3 pts each section, except part 1.5 which is 5 pts)*

1.1) Read the data into an object called `envdat` (do NOT use the option `as.is = TRUE`). Check the dimension to be sure the data loaded correctly. Then create a new object called `envdat2` which only contains information for the following countries : Austria, Iceland, Japan, New Zealand, Philippines, Russia, and Thailand. The variable that contains country is `country`. You'll need to use the ISO 3166 Standard Book of Country Codes which you can find HERE (https://en.wikipedia.org/wiki/ISO_3166-1_numeric#Officially_assigned_code_elements).

Check the dimensions of your results - you should have 9476 observations.

```
envdat <- read.csv("https://raw.githubusercontent.com/jreuning/sds230_data/refs/heads/main/ISS_20
20.csv")
dim(envdat)
```

```
## [1] 21718   224
```

```
country_list <- c("Austria", "Iceland", "Japan", "New Zealand", "Philippines", "Russia", "Thailan
d")
country_codes <- c(40, 352, 392, 554, 608, 643, 764)
envdat2 <- envdat[envdat$country %in% country_codes, ]
dim(envdat2)
```

```
## [1] 9476  224
```

1.2) Create a new variable called `Country` on `envdat2` which has Country names rather than Country numbers. There are several ways to do this, but I suggests you use the `recode()` function in the `car` package. The syntax for this function is something like

```
library(car)
NewVar <- recode(envdat2$country, "1 = 'Transylvania'; 2 = 'Bezerkystan';
                 3 = 'Isle Joyeuse'")
```

Once you're created the variable, make a table of the resulting variable to see how many observations there are from each country.

```
library(car)
envdat2$Country <- recode(envdat2$country,
                  "40 = 'Austria'; 352 = 'Iceland'; 392 = 'Japan';
                  554 = 'New Zealand'; 608 = 'Philippines'; 643 = 'Russia';
                  764 = 'Thailand'")
table(envdat2$Country)
```

```
##
##     Austria     Iceland       Japan New Zealand Philippines      Russia
##        1261        1150        1491         993        1500        1583
##    Thailand
##        1498
```

1.3) Make a variable `Gender` on `envdat2` that contains gender (which is variable `SEX`). Recode so that 1 becomes 'Male' and 2 becomes 'Female'. Again, make a table of resulting variable to see how many people identify as Male and how many as Female.

```
envdat2$Gender <- recode(envdat2$SEX,
                  "1 = 'Male'; 2 = 'Female'")
table(envdat2$Gender)
```

```
##
## Female   Male
##   5032   4444
```

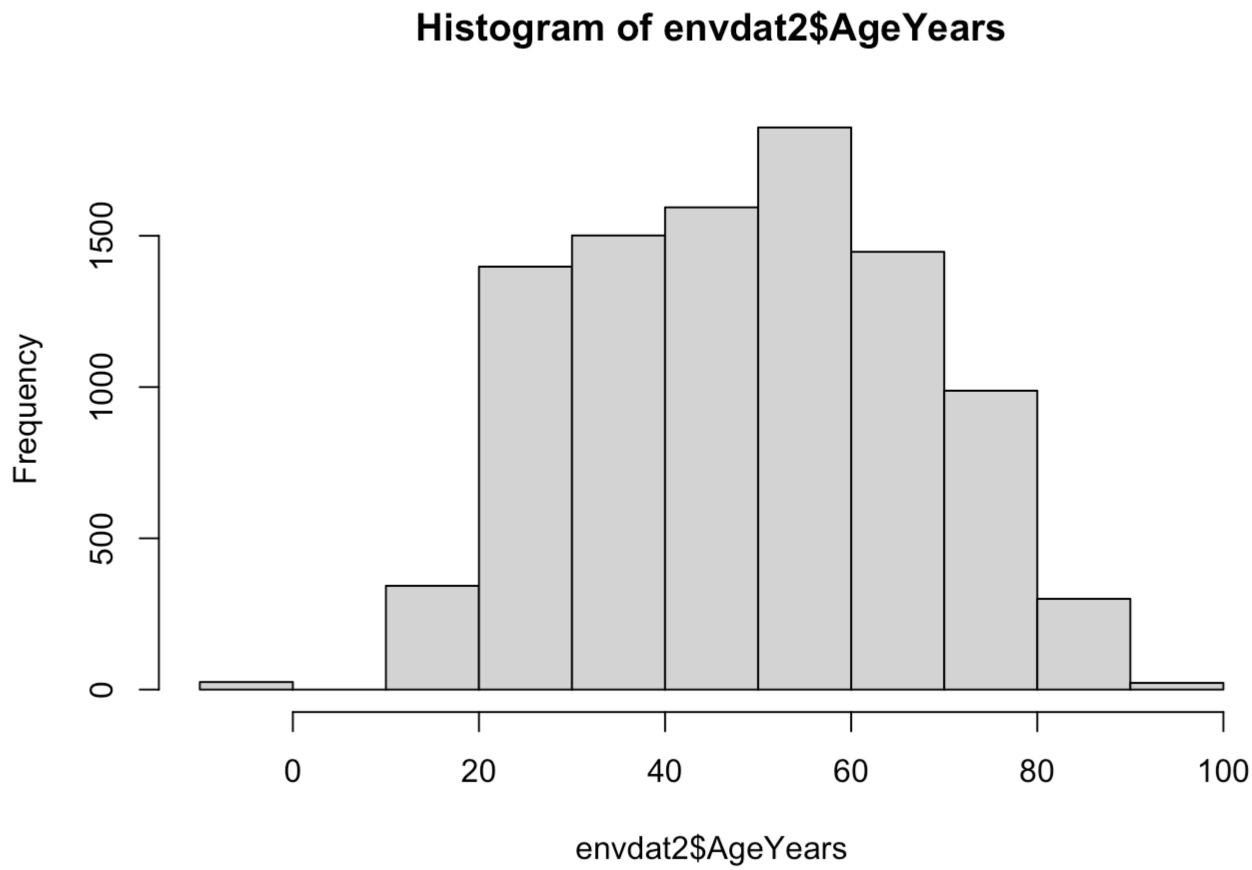
1.4) Create Variables `AgeYears` and `Educ` on `envdat2` from variables `AGE` and `EDUCYRS` which are age in years and years of education, respectively. Get summary information and make a histogram for each variable to see if values seem reasonable.

Modify both variables so that any negative values are changed to `NA`. You'll also discover that some people have more than 30 years of eduction. Assume this is an error and replace these values with NA. Repeat your histograms to make sure your code works.

```
envdat2$AgeYears <- envdat2$AGE
summary(envdat2$AgeYears)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   -9.00   35.00   50.00   49.35   63.00   97.00
```

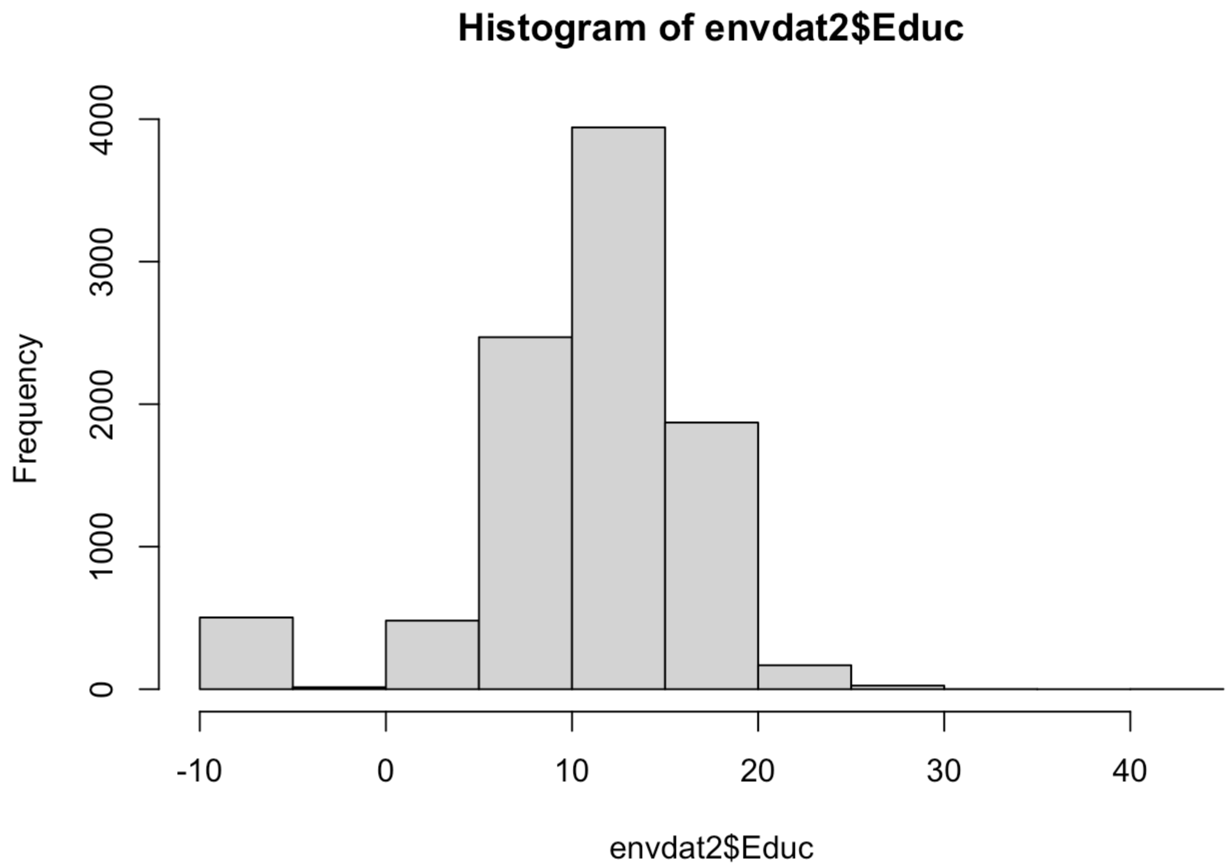
```
hist(envdat2$AgeYears)
```



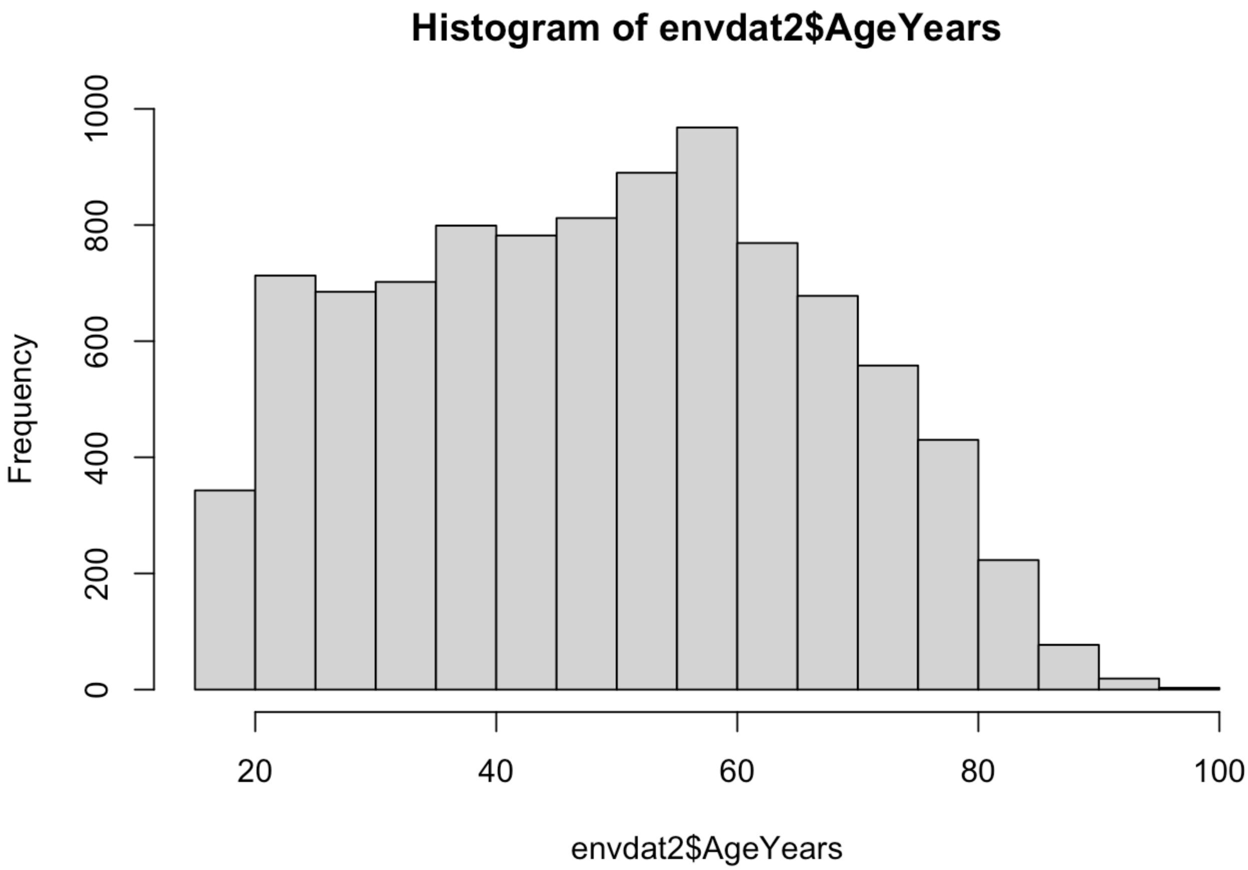

```
envdat2$Educ <- envdat2$EDUCYRS
summary(envdat2$Educ)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   -9.00   10.00   12.00   11.22   15.00   45.00
```

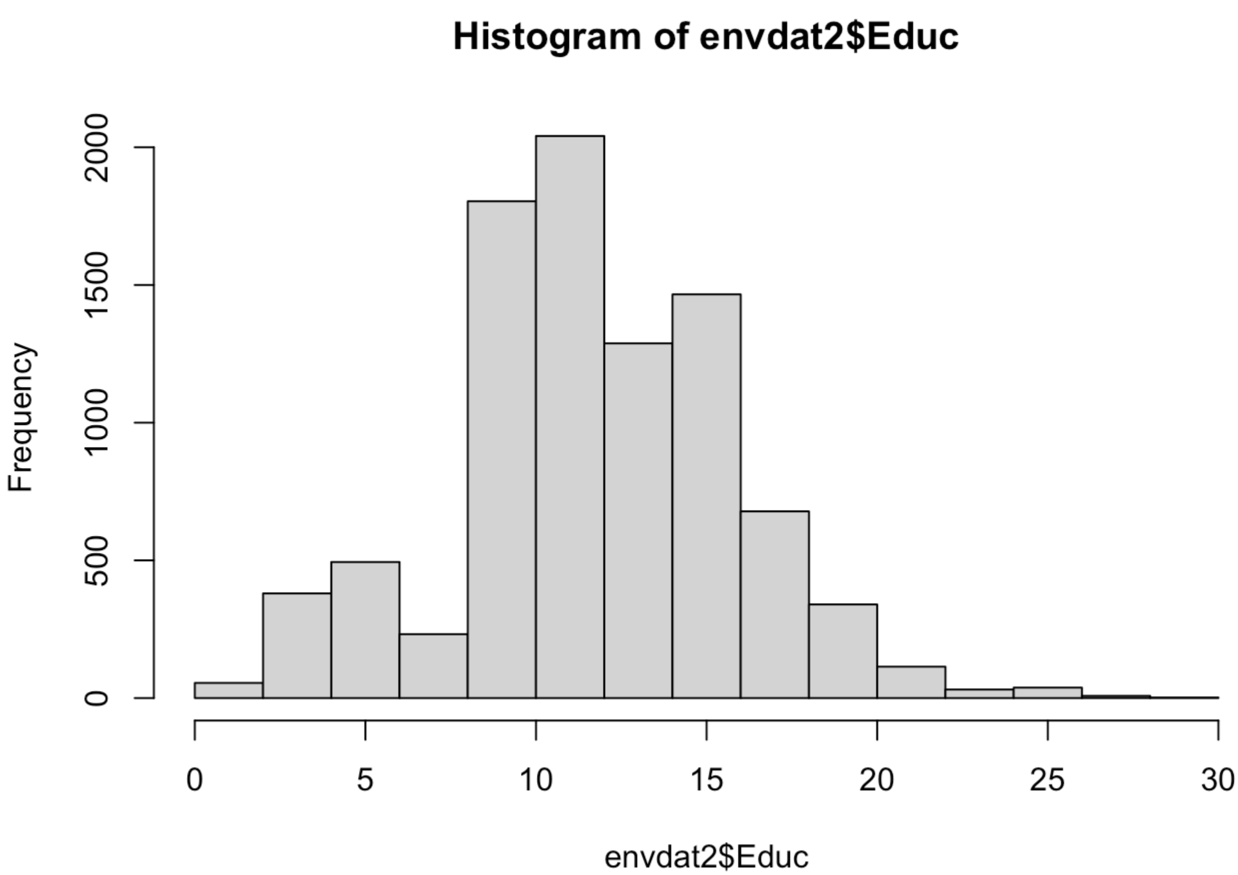
```
hist(envdat2$Educ)
```




```
envdat2$AgeYears[envdat2$AgeYears < 0] <- NA
envdat2$Educ[envdat2$Educ < 0 | envdat2$Educ > 30] <- NA

hist(envdat2$AgeYears)
```




```
hist(envdat2$Educ)
```



1.5) Make a variable `EmpStat` on `envdat2` from variable `MAINSTAT` that contains employment status information. The codebook for this question is listed below:

-9 = No answer -8 = Don't know 1 = In paid work 2 = Unemployed and looking for a job 3 = In education 4 = Apprentice or trainee 5 = Permanently sick or disabled 6 = Retired 7 = Domestic work (meaning working in the home with family) 8 = Incompulsory military service or community service 9 = Other

Code `EmpStat` in the following way : 1, 2, and 8 code as 'Working or Looking', 7 codes as 'At Home', 3 and 4 code as 'Student', 6 codes as 'Retired', -9 and -8 code as NA, and 5, 9 code as 'Other'.

Make a table of your results - look up options for the `table()` function so that it also displays NA values.

```
envdat2$EmpStat <- recode(envdat2$MAINSTAT,
                           "1 = 'Working or Looking'; 2 = 'Working or Looking'; 8 = 'Working or Lo
oking';
                           7 = 'At Home'; 3 = 'Student'; 4 = 'Student'; 6 = 'Retired'; -9 = NA; -8
= NA;
                           5 = 'Other'; 9 = 'Other'")
table(envdat2$EmpStat, useNA = "always")
```

```
##
##            At Home               Other             Retired             Student
##                919                 224                1557                 437
## Working or Looking                <NA>
##               5983                 356
```

1.6) If you look in the codebook starting on page 10, you'll notice many questions are on a 5 point scale. We're going to create a composite environmental index score from 12 of these questions. The new variable should be called `EnvAtt` on `envdat2`. It might be tempting to simply add scores together; however, for some questions 'Agree Strongly' (which is coded as 1) would indicate support the environment, while other questions 'Agree Strongly' indicates lack of support for the environment. Furthermore, you want to combined variables so that a higher score indicates more support for the environment.

The following questions should be included : V20, V21, V22, V23; V26, V27, V28, v29; V31, V32, V34, V36. For each quesion, you'll either want to add it directly, OR you'll want to add (6 - question). Using (6 - question) will convert 1's to 5's and 5's to 1's. Remember, for your final composite variable, you want a HIGHER score to indicate GREATER support for the environment.

FIRST - you'll need to replace the values -9 and -8 in each of these questions (which stand for 'No Answer' and 'Can't Choose') with NA. THEN you can proceed with making your composite variable.

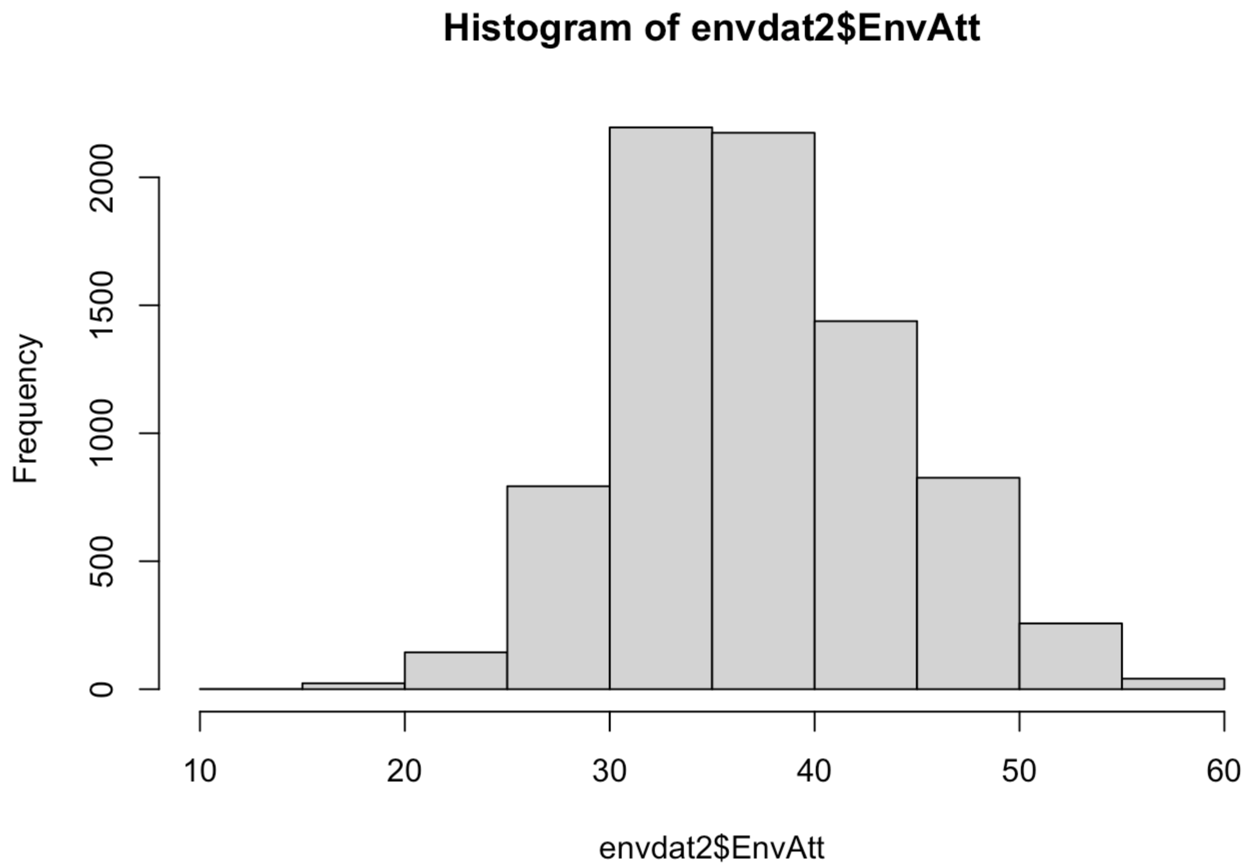
Your new varible should have a minimum of 12 and a maximum of 60. Check this by getting summary information and by making a histogram of `EnvAtt`. Also, make a normal quantile plot of `Envatt'. You'll also want to determine how many missing values were created in the final index.

The original variables were all on a 5 point scale and were certainly NOT normally distributed. Does your new composite variable seem to be approximately normally distributed?

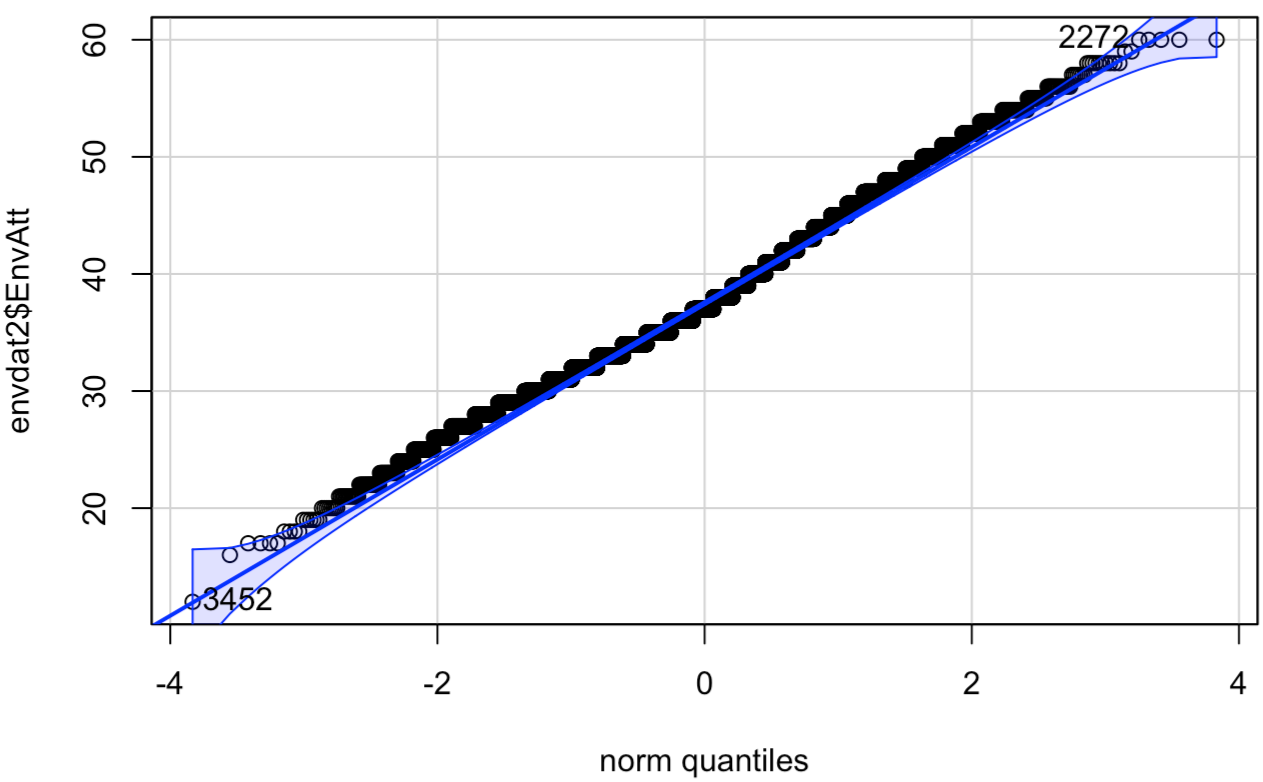
```r
# 1 = strongly agree, 5 = strongly disagree
#  Higher score = greater support for environment, so flip scale when needed
# v20 - Q10a Science will solve environmental problems (flip)
# v21 - Q10b Worry too much about environment and not enough about prices and jobs (flilp)
# v22 - Q10c Modern life harms the environment (flip)
# v23 - Q10d Worry too much about progress harming environment (flip)
# v26 - Q11a Protect environment: pay much higher prices (flip)
# v27 - Q11b Protect environment: pay much higher taxes (flip)
# v28 - Q11c Protect environment: cut your standard of living (flip)
# v29 - Q11d Protect environment: accept reduction of protected nature areas for economic develop
ment (keep)
# v31 - Q12b I do what is right, even when it costs more money and time (flip)
# v32 - Q12c There are more important things than protect environment (keep)
# v34 - Q12e Many claims about environment exaggerated (keep)
# v36 - Q12g Environmental problems have direct effect on everyday life (flip)

questions <- c("v20","v21","v22","v23","v26","v27","v28","v29","v31","v32","v34","v36")
envdat2[questions] <- lapply(envdat2[questions], function(x) {
  x[x %in% c(-9, -8)] <- NA
  x
})

envdat2$EnvAtt <- apply(envdat2[, c("v20", "v21", "v23", "v29", "v32", "v34")], 1, sum) + 36 - ap
ply(envdat2[, c("v22", "v26", "v27", "v28", "v31", "v36")], 1, sum)

summary(envdat2$EnvAtt)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's 
##   12.00   33.00   37.00   37.83   42.00   60.00    1584
```

```r
hist(envdat2$EnvAtt)
```




```
qqPlot(envdat2$EnvAtt)
```



```
## [1] 3452 2272
```

```
sum(is.na(envdat2$EnvAtt))
```

```
## [1] 1584
```

The new composite variable seems to be approximately normally distributed based on the qqplot and general shape of the histogram. Additionally, the directions said that we should have a minimum of 12 and a maximum of 60. The summary statistics show the correct maximum, but instead shows a minimum of 17, which I believe to be a reflection of the dataset rather than the true possible minimum (aka, there was no one who selected the most extreme option against environmental support for every question).

1.7) Finally, create a new dataframe called `envdat3` which contains only the new variables you've created in 1.2 through 1.6. Remove any rows with missing values for any of these variables. Get the dimension of `envdat3`. Lastly, attach `envdat3`.

```
new_vars <- c("Country", "Gender", "AgeYears", "Educ", "EmpStat", "EnvAtt")
envdat3 <- envdat2[, new_vars]
envdat3 <- na.omit(envdat3)
dim(envdat3)
```

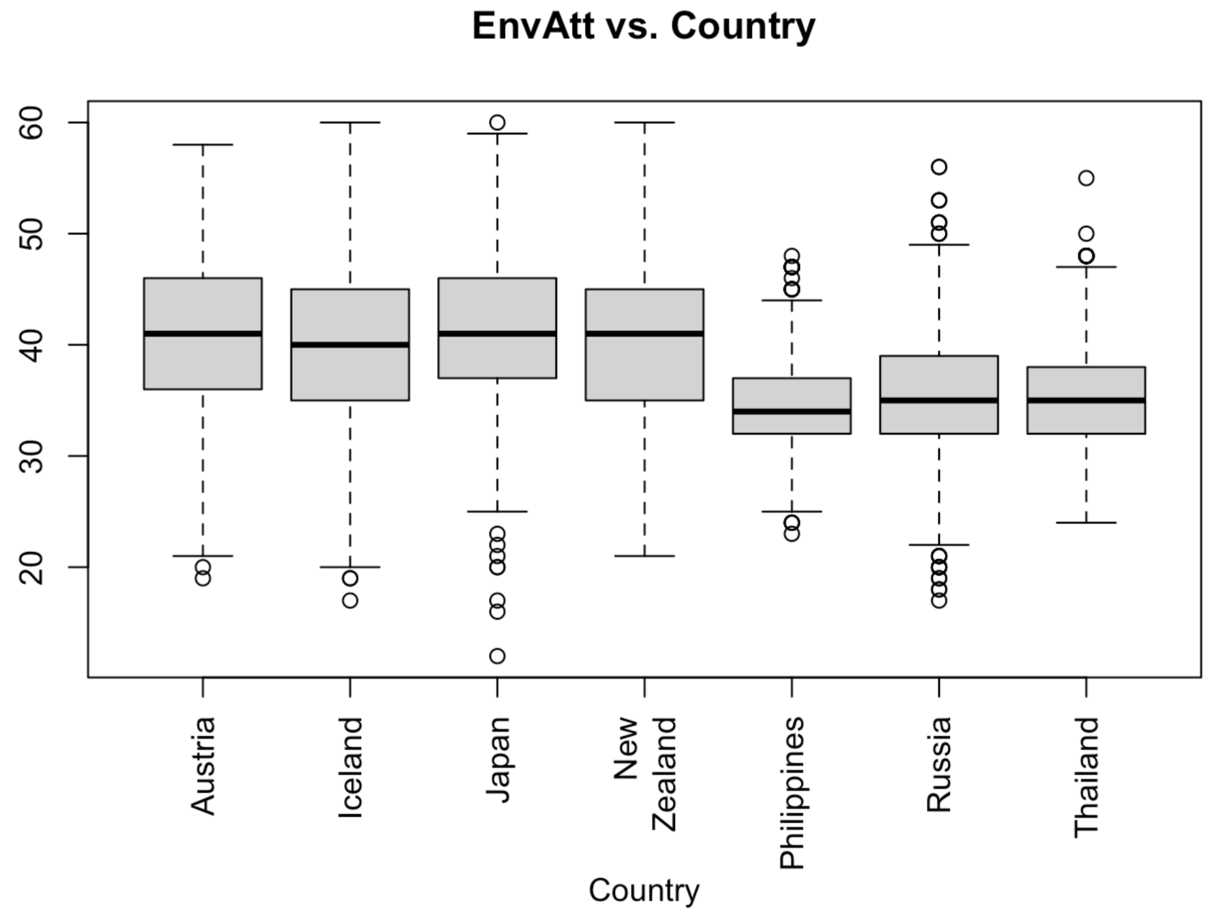
```
## [1] 7394    6
```

```
attach(envdat3)
```

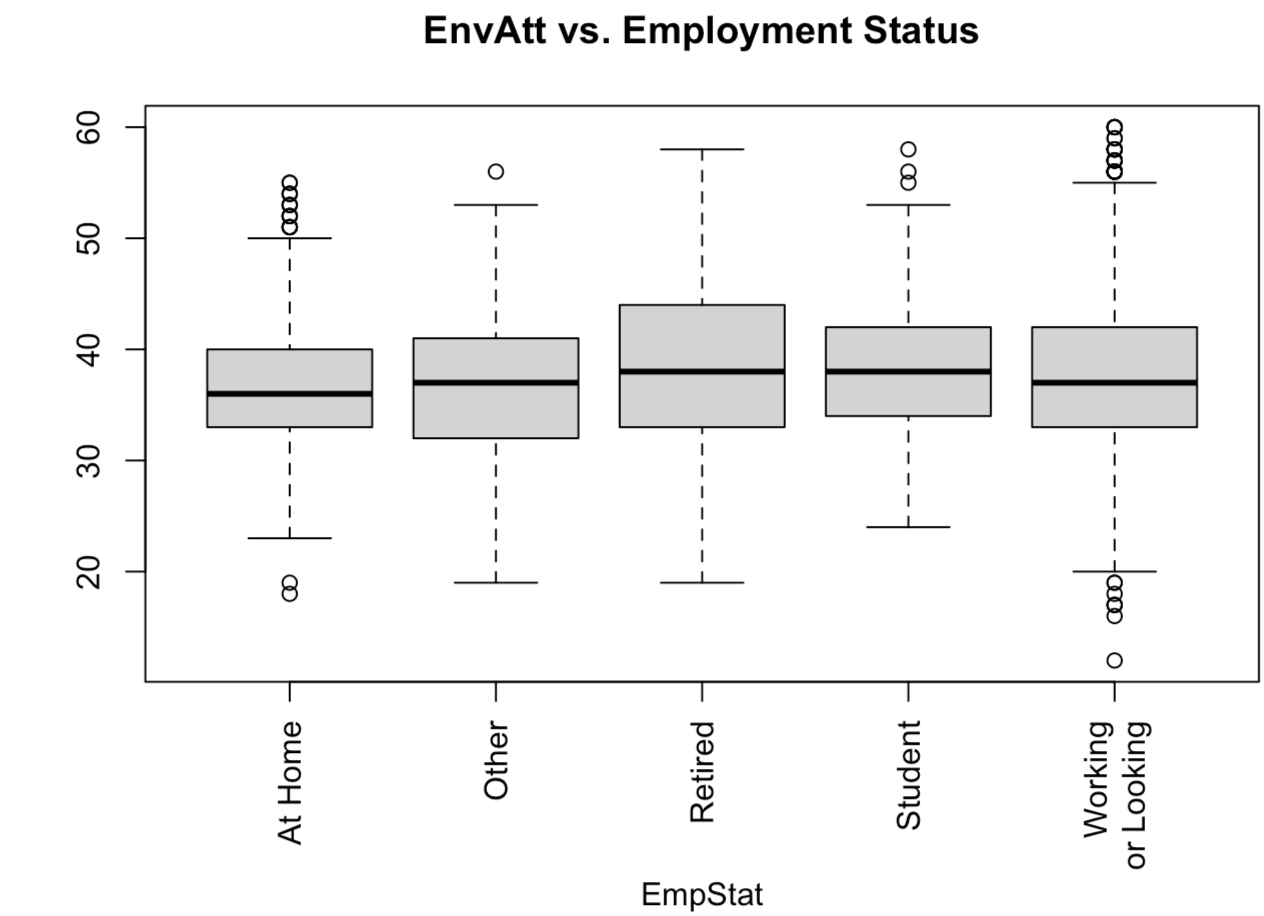
2. Plots and Interactions *(17 pts)*

2.1) Make boxplots of `EnvAtt` by Country, Employment Status, and Gender (that's 3 different boxplots). Additionally, make scatterplots of `EnvAtt` by age and by years of education. Is there evidence visually that our composite environmental index is related to any of these 5 variables (and elaborate)?

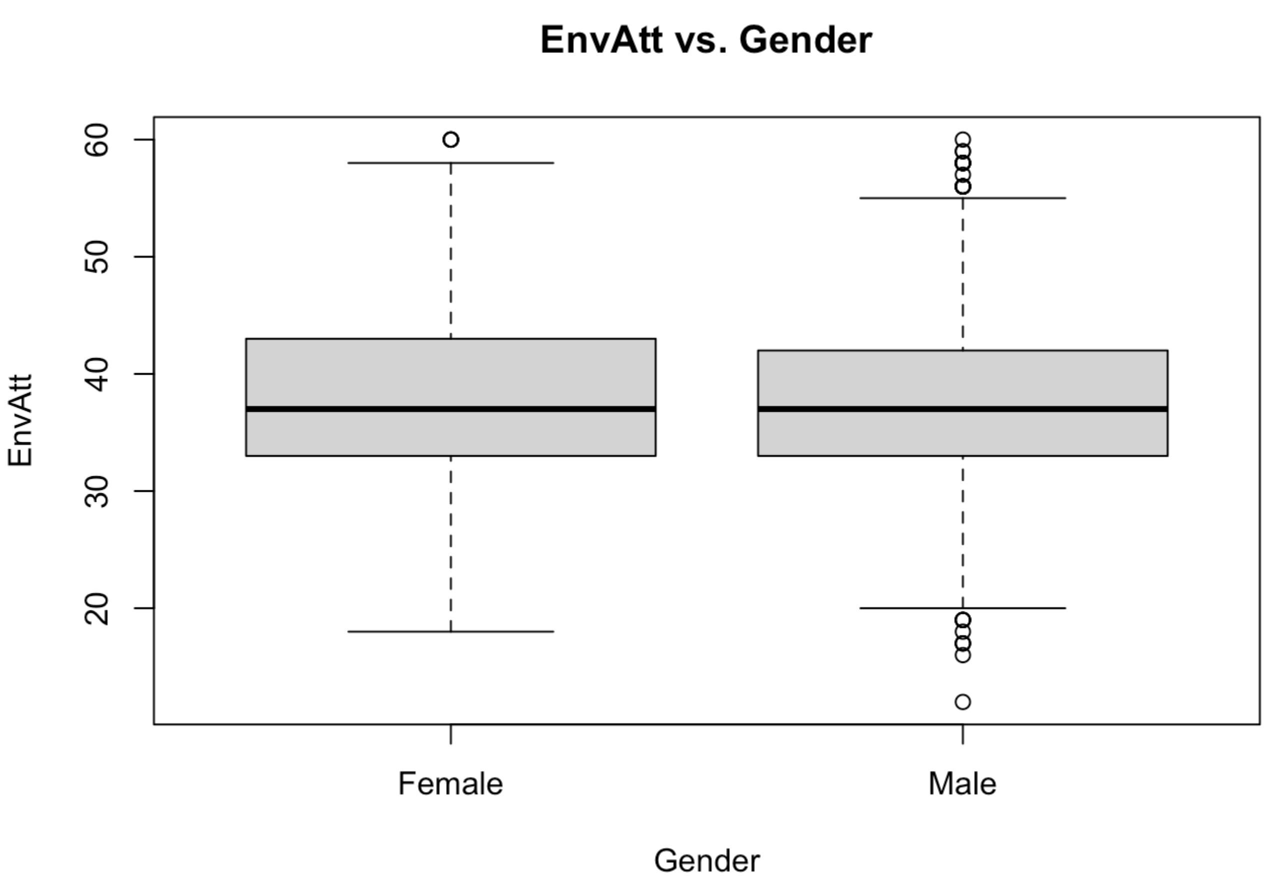
```
par(mar = c(6, 4, 4, 2), mgp = c(5, 1, 0))
boxplot(EnvAtt ~ Country, main = "EnvAtt vs. Country", ylab = "EnvAtt", xaxt = "n")
axis(1, at = 1:7,
     labels = c("Austria", "Iceland", "Japan", "New\nZealand", "Philippines", "Russia", "Thailan
d"), las = 2)
```



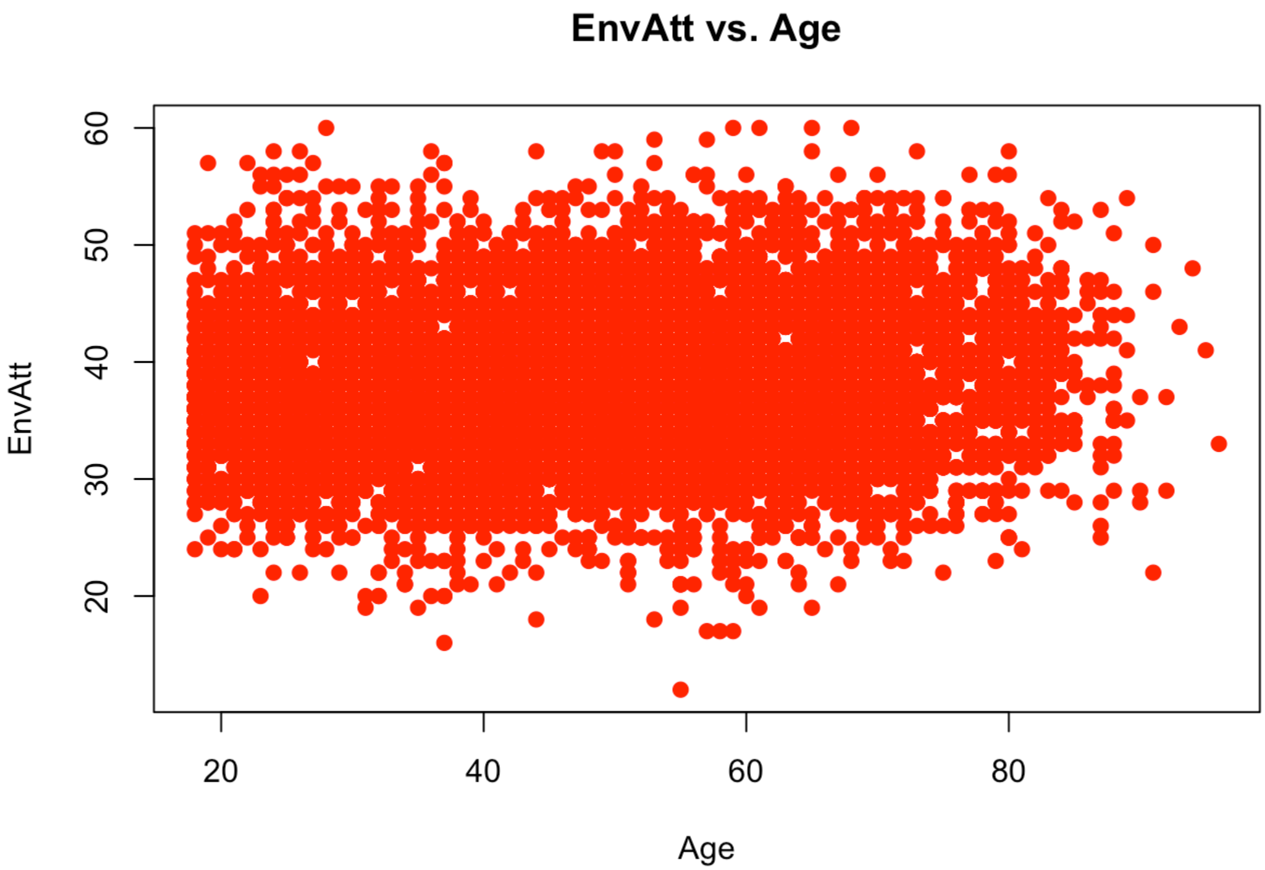

```
boxplot(EnvAtt ~ EmpStat, main = "EnvAtt vs. Employment Status", ylab = "EnvAtt", xaxt = "n")
axis(1, at = 1:5, labels = c("At Home", "Other", "Retired", "Student", "Working\nor Looking"), la
s = 2)
```



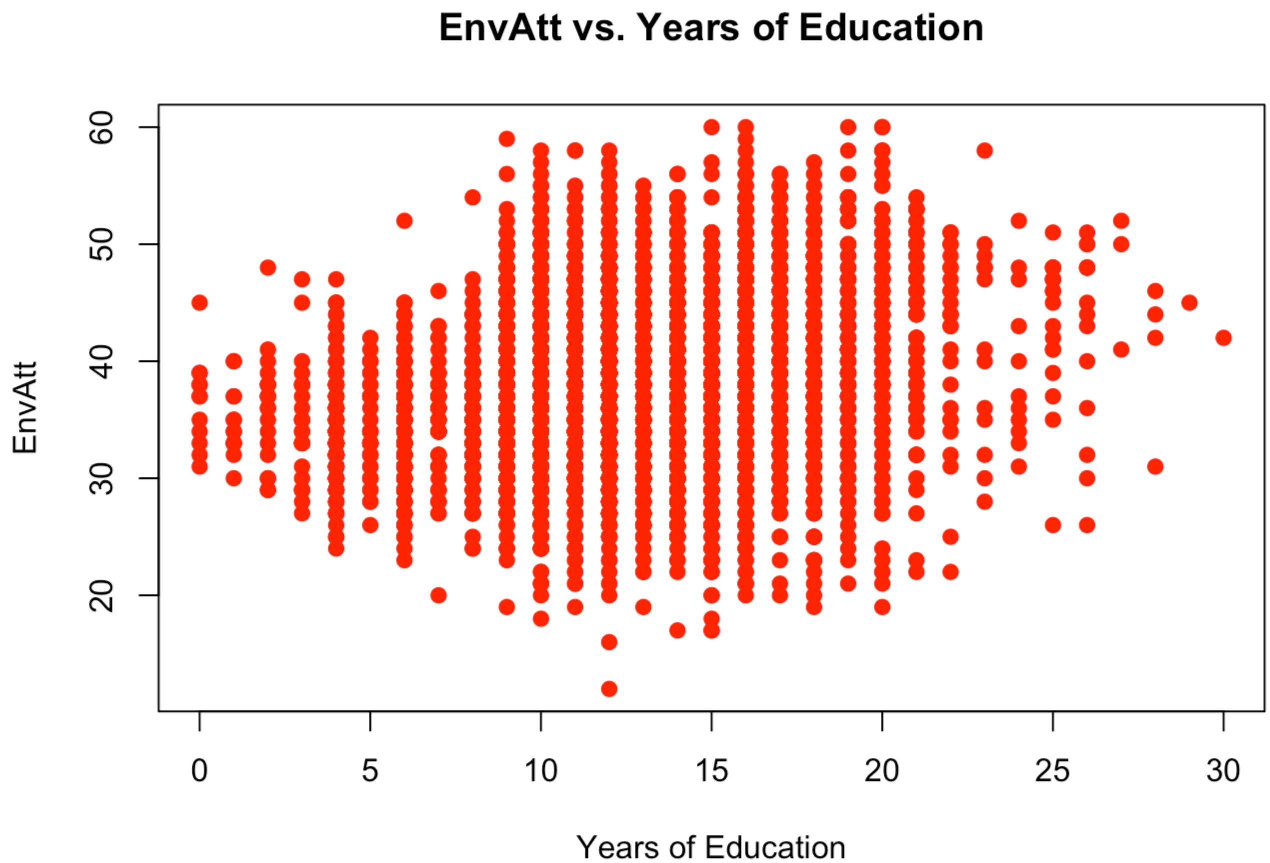

```
par(mar = c(5.1, 4.1, 4.1, 2.1), mgp = c(3, 1, 0))
boxplot(EnvAtt ~ Gender, main = "EnvAtt vs. Gender", ylab = "EnvAtt")
```



```
plot(EnvAtt ~ AgeYears, main = "EnvAtt vs. Age",
     xlab = "Age", ylab = "EnvAtt", pch = 19, col = "red")
```




```
plot(EnvAtt ~ Educ, main = "EnvAtt vs. Years of Education",
     xlab = "Years of Education", ylab = "EnvAtt", pch = 19, col = "red")
```



I do not think there is any visual evidence of the composite environmental index being related to any of the five variables. This is because in the boxplots, the medians and IQRs generally look very similar to each other, with the only small visual differences being the number of outliers. For the scatterplots, there were no obvious trends, as it seems like they averaged out to a similar score across all ages and years of education.

2.2). Examine relationships between the three possible pairs of the variables Gender, Country, EmpStat as they relate to EnvAtt. For each pair of categorical variables, make a table showing counts (i.e. `table(Gender, Country)` for example). This will allow you to see how many individuals exist for each combination of levels of your categorical variables. THEN, make interaction plots for each pair of variables as they relate to `EnvAtt`. Which pairs of categorical variables seem like they might have an interaction as they relate to `EnvAtt`?

```
table(Gender, Country)
```

```
##         Country
## Gender   Austria Iceland Japan New Zealand Philippines Russia Thailand
##   Female     544     393   471         417         688    725      583
##   Male       555     362   486         390         678    606      496
```

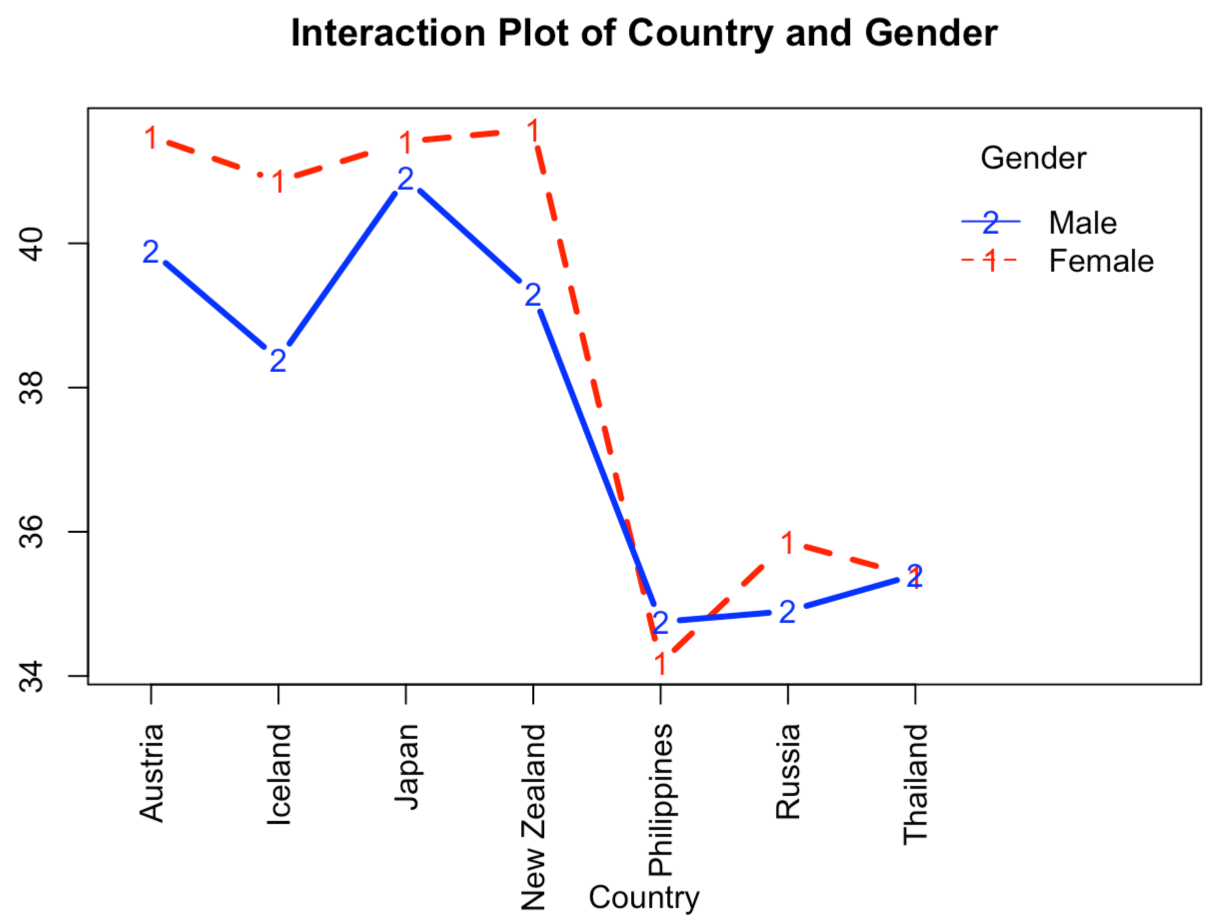
```
table(Gender, EmpStat)
```

```
##         EmpStat
## Gender   At Home Other Retired Student Working or Looking
##   Female     615    88     590     190               2338
##   Male        91    77     600     184               2621
```

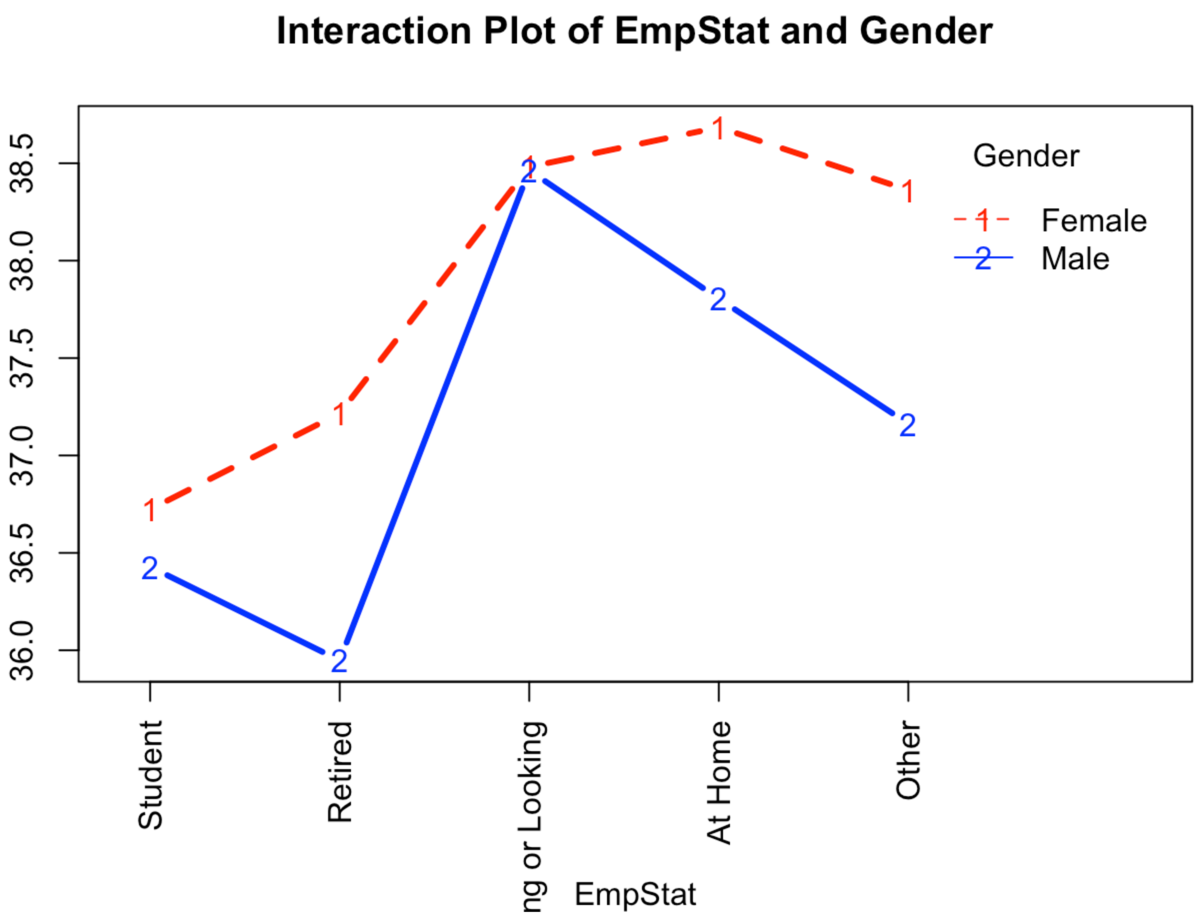
```
table(Country, EmpStat)
```

```
##                EmpStat
## Country        At Home Other Retired Student Working or Looking
##   Austria           22    27     405      35                610
##   Iceland            9    45      69      45                587
##   Japan            151    31     116      44                615
##   New Zealand       30     4     199      44                530
##   Philippines      303    21      81      82                879
##   Russia           113    32     255      52                879
##   Thailand          78     5      65      72                859
```

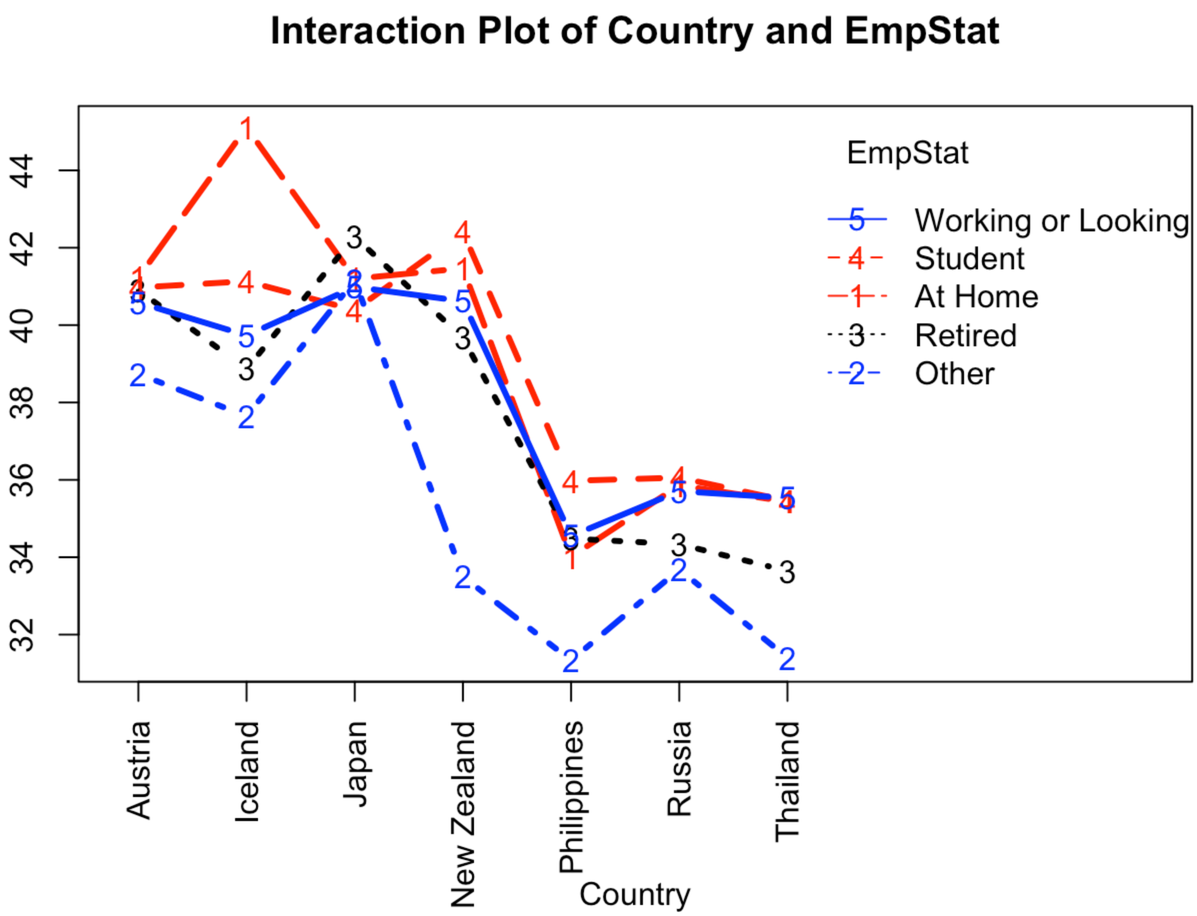
```
par(mar = c(6, 4, 4, 2), mgp = c(5, 1, 0))
CountryLabels <- factor(Country)
EmpStatLabels <- factor(EmpStat)
interaction.plot(Country, Gender, EnvAtt, type = 'b', lwd = 3, col = c('red', 'blue', 'black'),
                 main = "Interaction Plot of Country and Gender", xaxt = "n")
axis(1, at = 1:length(unique(Country)), labels = unique(CountryLabels), las = 2)
```




```
interaction.plot(EmpStat, Gender, EnvAtt, type = 'b', lwd = 3, col = c('red', 'blue', 'black'),
                 main = "Interaction Plot of EmpStat and Gender", xaxt = "n")
axis(1, at = 1:length(unique(EmpStatLabels)), labels = unique(EmpStatLabels), las = 2)
```




```
interaction.plot(Country, EmpStat, EnvAtt, type = 'b', lwd = 3, col = c('red', 'blue', 'black'),
                 main = "Interaction Plot of Country and EmpStat", xaxt = "n")
axis(1, at = 1:length(unique(Country)), labels = unique(CountryLabels), las = 2)
```




```
par(mar = c(5.1, 4.1, 4.1, 2.1), mgp = c(3, 1, 0))
```

There appears to be some interaction between Country and Gender and Employment Status and Country. There is some evidence of an interaction between Gender and Employment Status.

3. Fitting Two-Way ANOVA models *(20 pts, 10 pts each part)*

3.1) Fit three different two-way ANOVA models for `EnvAtt` , one for each pair of your three categorical variables (i.e. Country and EmpStat, Country and Gender, Gender and EmpStat). Include an interaction term in each model. Get summary information for each model using the `Anova()` function with option `type = 'III'` . Which models appear to have significant interaction terms? Does this seem to be consistent with what you observed in the interaction plots above?

```
Country_EmpStat_aov <- aov(EnvAtt ~ Country + EmpStat + Country*EmpStat)
Country_Gender_aov <- aov(EnvAtt ~ Country + Gender + Country*Gender)
Gender_EmpStat_aov <- aov(EnvAtt ~ Gender + EmpStat + Gender*EmpStat)

Anova(Country_EmpStat_aov, type = 'III')
```

```
## Anova Table (Type III tests)
##
## Response: EnvAtt
##                  Sum Sq   Df   F value    Pr(>F)
## (Intercept)      37393    1 1033.5723 < 2.2e-16 ***
## Country           7217    6   33.2475 < 2.2e-16 ***
## EmpStat            136    4    0.9425  0.438072
## Country:EmpStat   1719   24    1.9803  0.002955 **
## Residuals       266238 7359
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Anova(Country_Gender_aov, type = 'III')
```

```
## Anova Table (Type III tests)
##
## Response: EnvAtt
##                 Sum Sq   Df    F value    Pr(>F)
## (Intercept)    935991    1 26007.7321 < 2.2e-16 ***
## Country         37965    6   175.8169 < 2.2e-16 ***
## Gender            686    1    19.0727 1.275e-05 ***
## Country:Gender   1980    6     9.1676 5.048e-10 ***
## Residuals      265599 7380
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

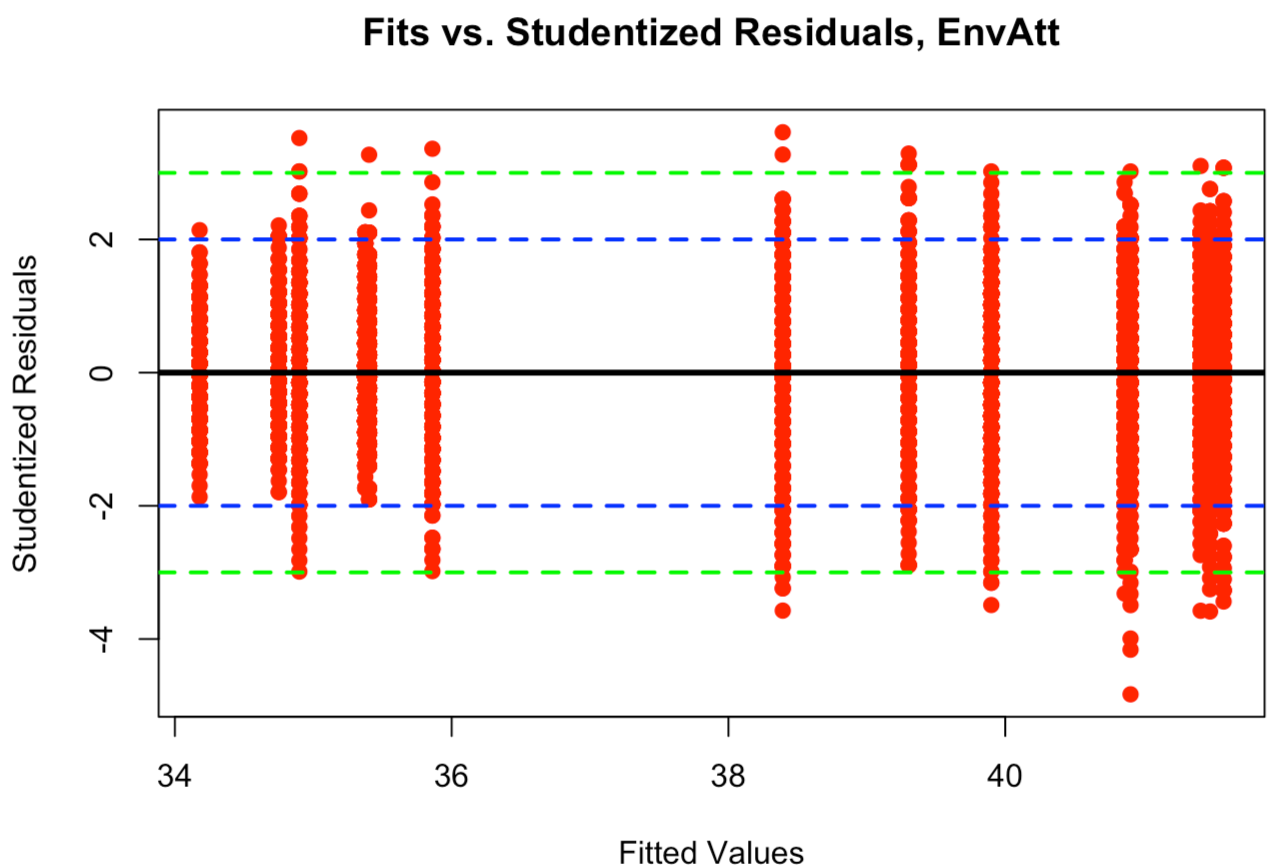
```
Anova(Gender_EmpStat_aov, type = 'III')
```

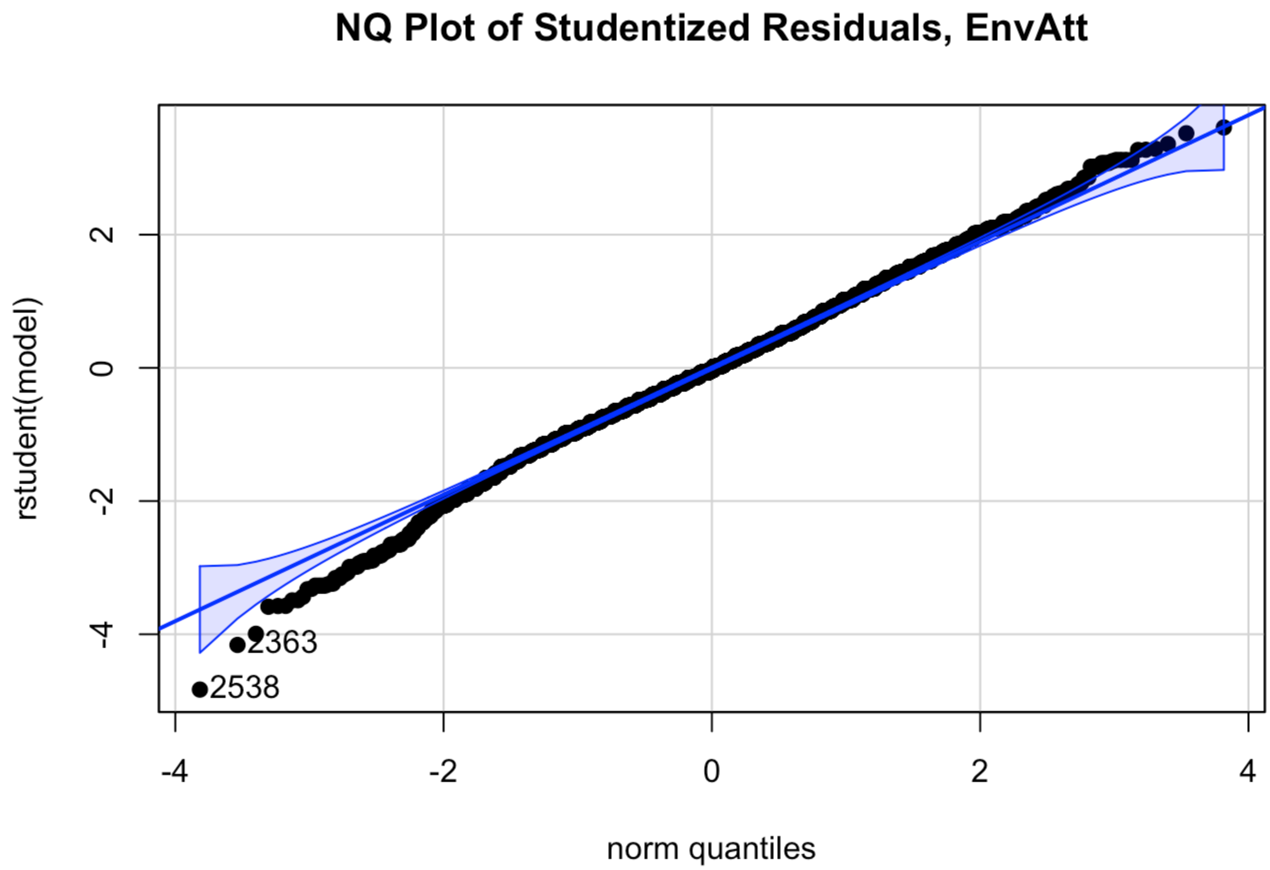
```
## Anova Table (Type III tests)
##
## Response: EnvAtt
##                 Sum Sq   Df    F value    Pr(>F)
## (Intercept)    829402    1 18981.5333 < 2.2e-16 ***
## Gender              7    1     0.1579   0.69112
## EmpStat          1545    4     8.8410  4.06e-07 ***
## Gender:EmpStat    374    4     2.1396   0.07325 .
## Residuals      322645 7384
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Country_EmpStat has a very significant interaction term (Pr(>F) = 8.092e-10) and Country_Gender also has a significant interaction term (Pr(>F) = 0.004516). This is consistent with what I had observed in the visual plots for Country_EmpStat (significant) and Gender_EmpStat (not significant), but not for Country_Gender (significant), though that may be because I do not quite know the exact point where I should visually discern significant vs insignificant interaction effects.

3.2) For the model with Country and Gender you fit in part (3.1), make residual plots. Are the model assumptions reasonably met (write a sentence supporting your answer)?

```r
myResPlots <- function(model, label){
  #plot of fitted vs. studentized residuals
  plot(rstudent(model) ~ model$fitted.values, pch = 19, col = 'red',
       xlab = "Fitted Values", ylab = "Studentized Residuals",
       main = paste("Fits vs. Studentized Residuals,", label))
  abline(h = 0, lwd = 3)
  abline(h = c(2,-2), lty = 2, lwd = 2, col="blue")
  abline(h = c(3,-3), lty = 2, lwd = 2, col="green")

  #Normal quantile plot of studentized residuals
  qqp(rstudent(model), pch = 19, main = paste("NQ Plot of Studentized Residuals,", label))

  # #Plot of Cooks Distances
  # plot(model, which = 4, lwd = 3, col = 'red')
  # abline(h = 6/length(model$residuals), col = 'black', lwd = 3, lty = 2)
}

myResPlots(Country_Gender_aov, label = "EnvAtt")
```






```
## [1] 2538 2363
```

Model assumptions seem to be reasonably met. Normality is good when looking at the qqplot, as the point generally stay within the boundaries, and the fits vs residuals plot shows equal variances.

4. ANCOVA *(20 pts, 10 pts each part)*

4.1) Fit an ANCOVA model predicting `EnvAtt` based on Gender, Education, and the interaction of Gender and Education. Get ANOVA summary information for this model (again, use the `Anova()` function with option `type = 'III'`). Is there a significant interaction between Gender and Education? Also get linear model summary information for this model. For which gender is there a greater increase in Environmental Index Score as Education increases?

```
Gender_Educ_model <- lm(EnvAtt ~ Gender*Educ)
Anova(Gender_Educ_model, type = 'III')
```

```
## Anova Table (Type III tests)
##
## Response: EnvAtt
##             Sum Sq   Df  F value    Pr(>F)
## (Intercept) 350402    1 8508.628 < 2.2e-16 ***
## Gender         604    1   14.661 0.0001298 ***
## Educ         15574    1  378.170 < 2.2e-16 ***
## Gender:Educ   1194    1   28.996 7.475e-08 ***
## Residuals   304335 7390
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

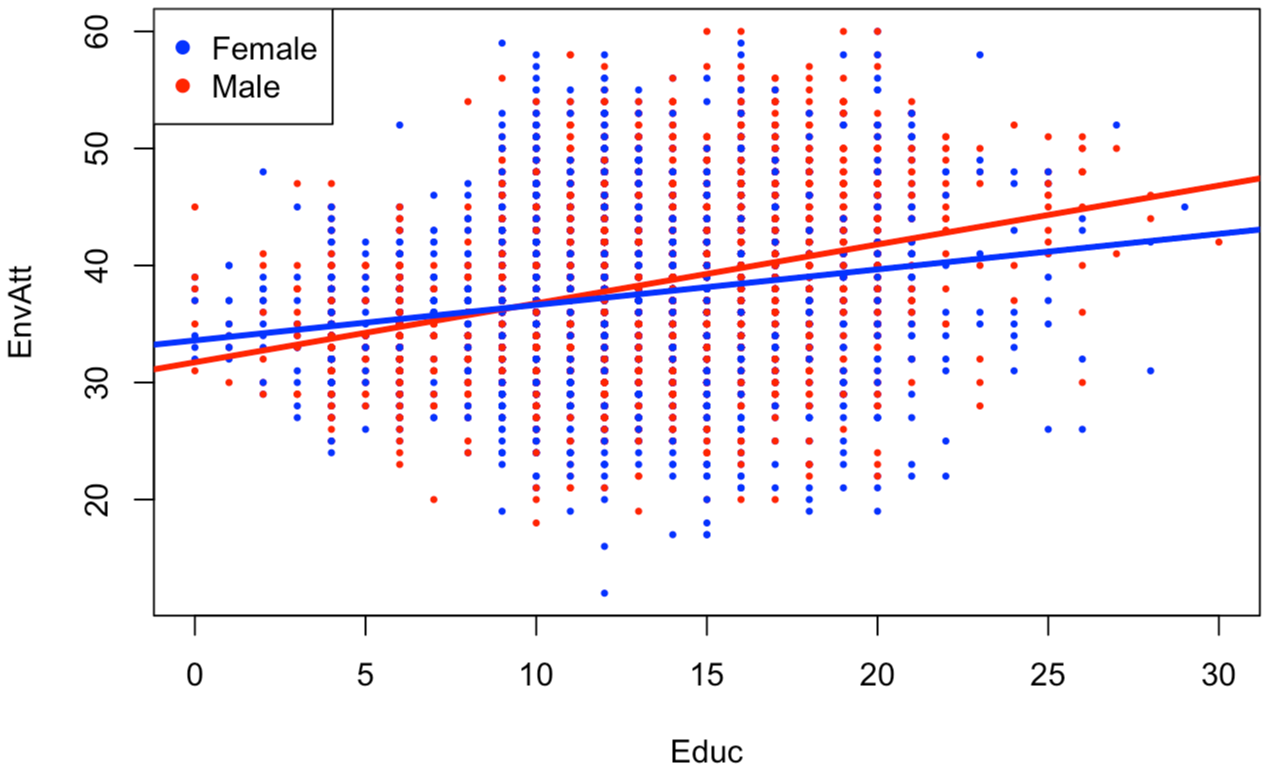
```
summary(Gender_Educ_model)
```

```
##
## Call:
## lm(formula = EnvAtt ~ Gender * Educ)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -25.2458  -4.2849  -0.4226   4.2188  22.6658
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)     31.72925    0.34398  92.242  < 2e-16 ***
## GenderMale       1.87018    0.48843   3.829  0.00013 ***
## Educ             0.50371    0.02590  19.447  < 2e-16 ***
## GenderMale:Educ -0.19984    0.03711  -5.385 7.48e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.417 on 7390 degrees of freedom
## Multiple R-squared:  0.0673, Adjusted R-squared:  0.06692
## F-statistic: 177.8 on 3 and 7390 DF,  p-value: < 2.2e-16
```

There is a significant interaction between Gender and Education (Pr(>F) = 0.004365). Because of the negative value for GenderMale:Educ, the model suggests that there is a greater increase in EnvAtt as Education increases among females.

4.2) Make a plot that shows `EnvAtt` as predicted by years of education with separate colors for each gender. Then, superimpose the two predicted regression lines (one for each gender). The plot should be consistent with your results from part a). You'll want to look carefully at similar code in Class 19.

```
plot(EnvAtt ~ Educ, col = ifelse(Gender == "Male", "blue", "red"), pch = 16, cex = 0.5)
legend("topleft", col = c("blue", "red"), legend = levels(factor(Gender)), pch = 16)
coefs <- coef(Gender_Educ_model)
abline(a = coefs["(Intercept)"], b = coefs["Educ"], col = "red", lwd = 3)
abline(a = coefs["(Intercept)"] + coefs["GenderMale"],
       b = coefs["Educ"] + coefs["GenderMale:Educ"], col = "blue", lwd = 3)
```



5. GLM *(20 pts, 10 pts each part)*

5.1). Fit a model for `EnvAtt` that includes ALL of the five possible continous and categorical predictors. Also include two-way interactions between Gender and Education, Employment Status and Country, Gender and Country, Employment Status and Gender, Age and Gender. Save to an object called `m1`.

Get ANOVA summary information for this model (again, use `Anova()` with option `type = 'III'`).

THEN, perform manual backwards stepwise regression, removing non-significant terms until all terms have p-values less than 0.05. REMEMBER, you want to remove interactions BEFORE you remove any main effects. You don't need to show every step - just put your final model into an object called `m2`. Get linear model summary information for this model. Finally, check residuals for your final model.

```
m1 <- lm(EnvAtt ~ Gender + Country + EmpStat + AgeYears + Educ + Gender*Educ
         + EmpStat*Country + Gender*Country + EmpStat*Gender + AgeYears*Gender)
Anova(m1, type = 'III')
```

```
## Anova Table (Type III tests)
##
## Response: EnvAtt
##                   Sum Sq   Df  F value    Pr(>F)
## (Intercept)      23211    1 663.2414 < 2.2e-16 ***
## Gender               3    1   0.0875 0.7674492
## Country           4452    6  21.2049 < 2.2e-16 ***
## EmpStat            177    4   1.2651 0.2812535
## AgeYears            70    1   2.0065 0.1566664
## Educ              4235    1 121.0026 < 2.2e-16 ***
## Gender:Educ        417    1  11.9180 0.0005591 ***
## Country:EmpStat   1452   24   1.7289 0.0149056 *
## Gender:Country     913    6   4.3503 0.0002168 ***
## Gender:EmpStat      85    4   0.6078 0.6569962
## Gender:AgeYears      7    1   0.1920 0.6612754
## Residuals       257008 7344
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
m2 <- lm(EnvAtt ~ Gender + Country + EmpStat + Educ + Gender*Educ + EmpStat*Country + Gender*Coun
try)
Anova(m2, type = "III")
```

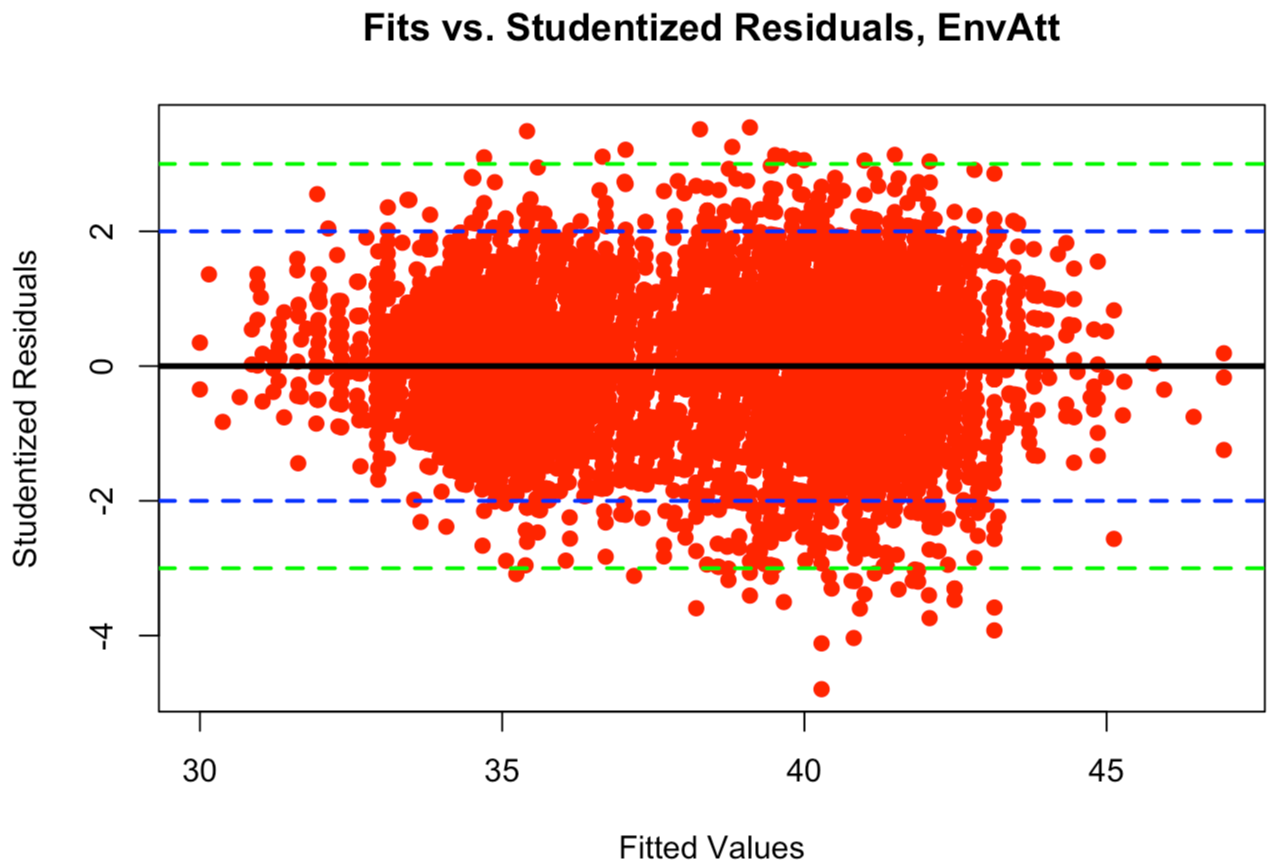
```
## Anova Table (Type III tests)
##
## Response: EnvAtt
##                   Sum Sq   Df  F value    Pr(>F)
## (Intercept)      29487    1 842.6723 < 2.2e-16 ***
## Gender               2    1   0.0455 0.8311805
## Country           5242    6  24.9670 < 2.2e-16 ***
## EmpStat            188    4   1.3414 0.2519121
## Educ              4483    1 128.1198 < 2.2e-16 ***
## Gender:Educ        494    1  14.1152 0.0001733 ***
## Country:EmpStat   1554   24   1.8500 0.0069771 **
## Gender:Country     898    6   4.2783 0.0002606 ***
## Residuals       257190 7350
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

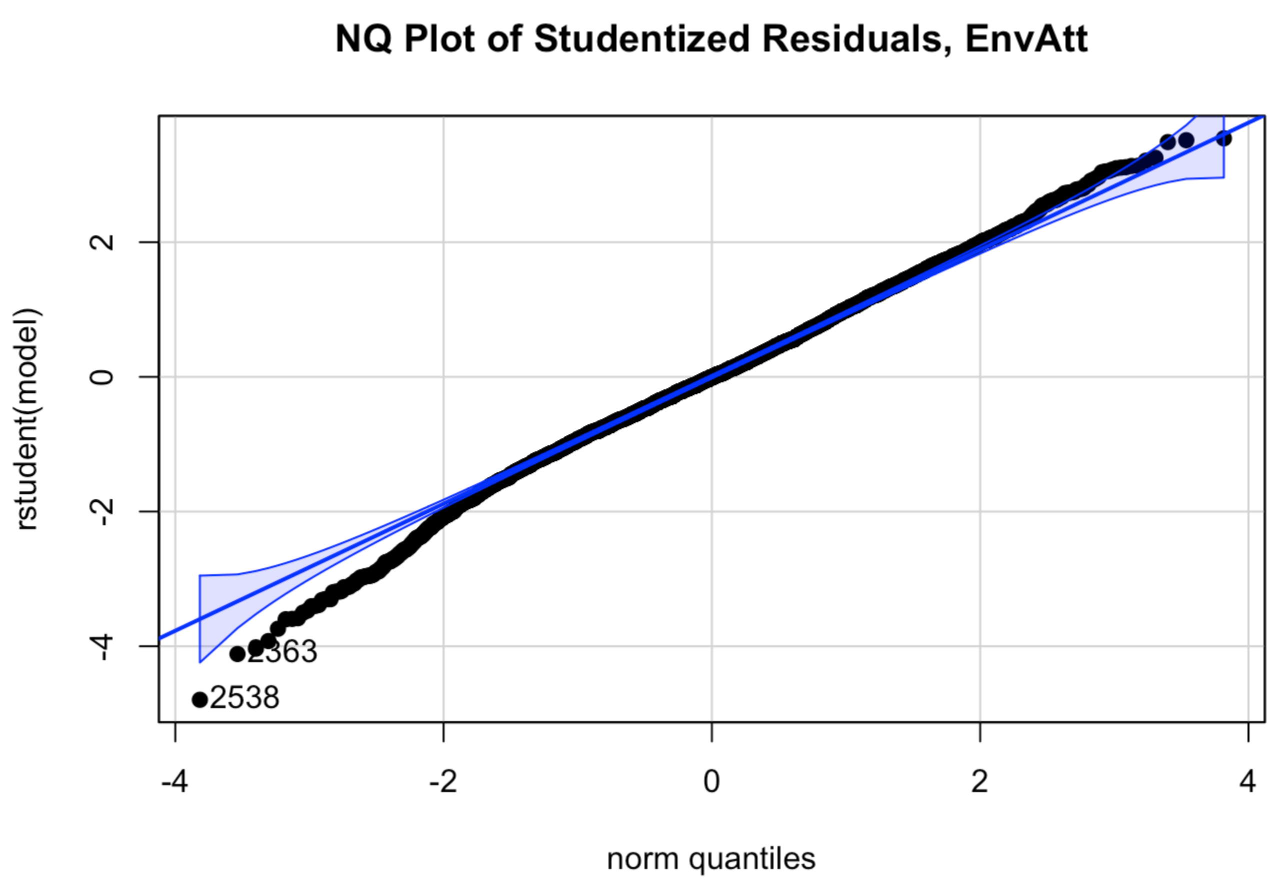
```
summary(m2)
```

```
##
## Call:
## lm(formula = EnvAtt ~ Gender + Country + EmpStat + Educ + Gender *
##     Educ + EmpStat * Country + Gender * Country)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -28.2838  -3.7339  -0.0315   3.7725  20.9013
##
## Coefficients:
##                                              Estimate Std. Error t value
## (Intercept)                                 37.706926   1.298948  29.029
## GenderMale                                   0.127674   0.598859   0.213
## CountryIceland                               3.300957   2.355106   1.402
## CountryJapan                                -0.605495   1.351492  -0.448
## CountryNew Zealand                          -0.907834   1.667581  -0.544
## CountryPhilippines                          -6.735039   1.307686  -5.150
## CountryRussia                               -5.963427   1.382558  -4.313
## CountryThailand                             -5.217981   1.429540  -3.650
## EmpStatOther                                -2.323669   1.701006  -1.366
## EmpStatRetired                               0.226034   1.306504   0.173
## EmpStatStudent                               0.474699   1.630802   0.291
## EmpStatWorking or Looking                   -0.167041   1.299529  -0.129
## Educ                                         0.329564   0.029116  11.319
## GenderMale:Educ                             -0.151873   0.040424  -3.757
## CountryIceland:EmpStatOther                 -4.955253   2.749852  -1.802
## CountryJapan:EmpStatOther                    2.654208   2.073359   1.280
## CountryNew Zealand:EmpStatOther             -3.584369   3.596212  -0.997
## CountryPhilippines:EmpStatOther             -0.626657   2.172746  -0.288
## CountryRussia:EmpStatOther                   0.303754   2.073750   0.146
## CountryThailand:EmpStatOther                -1.105699   3.221452  -0.343
## CountryIceland:EmpStatRetired               -5.965260   2.471088  -2.414
## CountryJapan:EmpStatRetired                  1.195620   1.524454   0.784
## CountryNew Zealand:EmpStatRetired           -0.528991   1.755272  -0.301
## CountryPhilippines:EmpStatRetired           -0.189294   1.515979  -0.125
## CountryRussia:EmpStatRetired                -1.614991   1.467928  -1.100
## CountryThailand:EmpStatRetired              -1.696195   1.651969  -1.027
## CountryIceland:EmpStatStudent               -5.111654   2.707488  -1.888
## CountryJapan:EmpStatStudent                 -1.142221   1.930822  -0.592
## CountryNew Zealand:EmpStatStudent            1.126119   2.155986   0.522
## CountryPhilippines:EmpStatStudent            0.376934   1.794465   0.210
## CountryRussia:EmpStatStudent                 0.062844   1.909744   0.033
## CountryThailand:EmpStatStudent              -2.014445   1.901588  -1.059
## CountryIceland:EmpStatWorking or Looking    -5.551059   2.374535  -2.338
## CountryJapan:EmpStatWorking or Looking      -0.136706   1.420224  -0.096
## CountryNew Zealand:EmpStatWorking or Looking -0.079605   1.713897  -0.046
## CountryPhilippines:EmpStatWorking or Looking  0.146312   1.368005   0.107
## CountryRussia:EmpStatWorking or Looking      0.192300   1.431128   0.134
## CountryThailand:EmpStatWorking or Looking   -0.536434   1.481178  -0.362
## GenderMale:CountryIceland                    0.127378   0.595785   0.214
## GenderMale:CountryJapan                      1.226117   0.559254   2.192
## GenderMale:CountryNew Zealand               -0.006406   0.570642  -0.011
## GenderMale:CountryPhilippines                1.999056   0.512325   3.902
## GenderMale:CountryRussia                     0.849649   0.499080   1.702
## GenderMale:CountryThailand                   1.726795   0.519496   3.324
##                                             Pr(>|t|)
```

```
## (Intercept)                                  < 2e-16 ***
## GenderMale                                  0.831181
## CountryIceland                              0.161072
## CountryJapan                                0.654152
## CountryNew Zealand                          0.586181
## CountryPhilippines                          2.67e-07 ***
## CountryRussia                               1.63e-05 ***
## CountryThailand                             0.000264 ***
## EmpStatOther                                0.171963
## EmpStatRetired                              0.862651
## EmpStatStudent                              0.770996
## EmpStatWorking or Looking                   0.897725
## Educ                                         < 2e-16 ***
## GenderMale:Educ                             0.000173 ***
## CountryIceland:EmpStatOther                 0.071585 .
## CountryJapan:EmpStatOther                   0.200533
## CountryNew Zealand:EmpStatOther             0.318940
## CountryPhilippines:EmpStatOther             0.773036
## CountryRussia:EmpStatOther                  0.883550
## CountryThailand:EmpStatOther                0.731435
## CountryIceland:EmpStatRetired               0.015802 *
## CountryJapan:EmpStatRetired                 0.432893
## CountryNew Zealand:EmpStatRetired           0.763139
## CountryPhilippines:EmpStatRetired           0.900633
## CountryRussia:EmpStatRetired                0.271288
## CountryThailand:EmpStatRetired              0.304562
## CountryIceland:EmpStatStudent               0.059069 .
## CountryJapan:EmpStatStudent                 0.554155
## CountryNew Zealand:EmpStatStudent           0.601462
## CountryPhilippines:EmpStatStudent           0.833632
## CountryRussia:EmpStatStudent                0.973750
## CountryThailand:EmpStatStudent              0.289476
## CountryIceland:EmpStatWorking or Looking    0.019427 *
## CountryJapan:EmpStatWorking or Looking      0.923319
## CountryNew Zealand:EmpStatWorking or Looking 0.962955
## CountryPhilippines:EmpStatWorking or Looking 0.914830
## CountryRussia:EmpStatWorking or Looking     0.893114
## CountryThailand:EmpStatWorking or Looking   0.717238
## GenderMale:CountryIceland                   0.830710
## GenderMale:CountryJapan                     0.028381 *
## GenderMale:CountryNew Zealand               0.991044
## GenderMale:CountryPhilippines               9.63e-05 ***
## GenderMale:CountryRussia                    0.088717 .
## GenderMale:CountryThailand                  0.000892 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.915 on 7350 degrees of freedom
## Multiple R-squared:  0.2118, Adjusted R-squared:  0.2072
## F-statistic: 45.93 on 43 and 7350 DF,  p-value: < 2.2e-16
```

```
myResPlots(m2, label = "EnvAtt")
```






```
## [1] 2538 2363
```

5.2). Write a few sentences describing the overall fit of your final model, the direction of coefficients for each continuous predictor, some discussion of categorical predictors, and the nature of any resulting interactions.

The final model predicting the composite environmental attitude scale included Gender, Country, Employment Status, Education level, and interactions between Gender and Education, Gender and Country, and Country and Employment Status. Overall scores were highest in Iceland and lowest in Phillipines. Overall, scores were similar between gender groups. However, whilel more years of education was overall associated with higher scores, the effect was less prominent for males. Students had the highest scores overall, while those listed as “Other” had the lowest scores. There was an interaction between Country and Gender, as well as Country and Employment Status. The overall R-squared was 21%. The fit of hte model was good with the residuals being approximately normally distributed and there was no evidence of non-linear trends, heteroskedasticity, outliers, or influential points.

THE END